Assignment 2

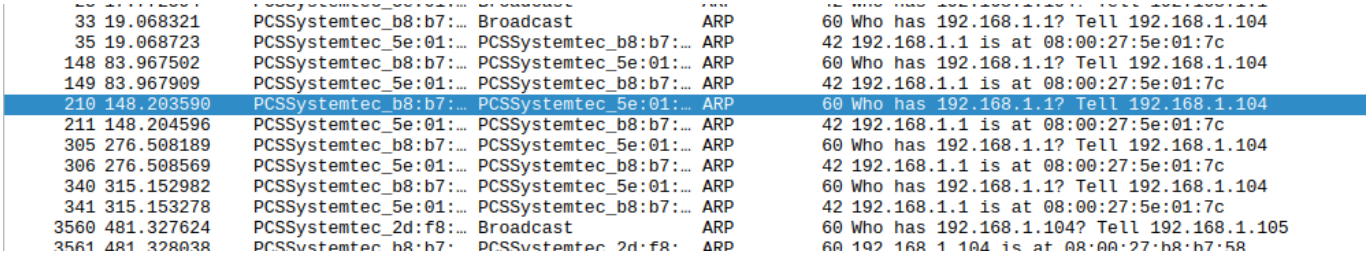
```
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Course : MCA
Subject : Data Communications and Computer Networks
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Wireshark Analysis

Task 1: Analyze an ARP Request and Reply

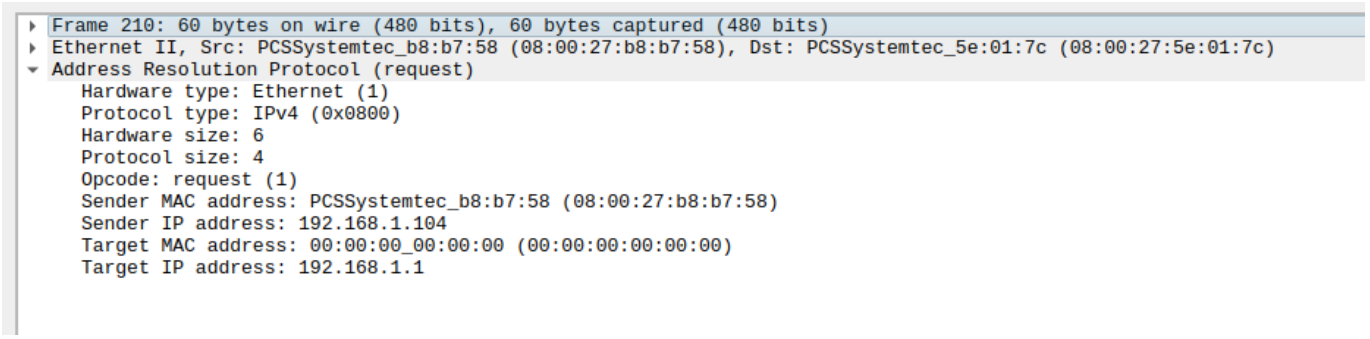
Identify one complete ARP request–reply pair in the provided capture. Use the same request–reply pair for all questions in this task.

arp.opcode == 1 || arp.opcode == 2



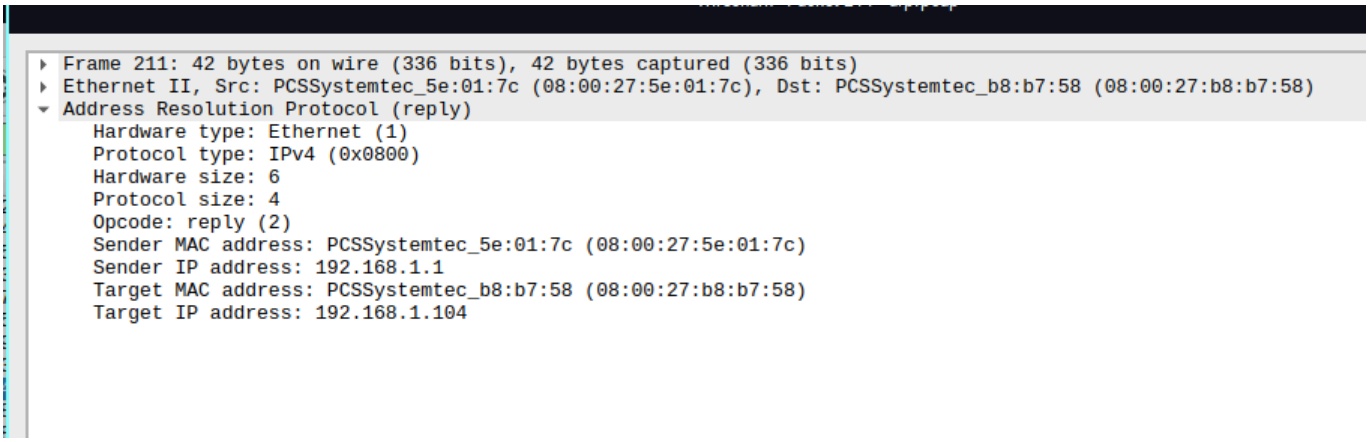
33	19.068321	PCSSystemtec_b8:b7:…	Broadcast	ARP	60 Who has 192.168.1.1? Tell 192.168.1.104
35	19.068723	PCSSystemtec_5e:01:…	PCSSystemtec_b8:b7:…	ARP	42 192.168.1.1 is at 08:00:27:5e:01:7c
148	83.967502	PCSSystemtec_b8:b7:…	PCSSystemtec_5e:01:…	ARP	60 Who has 192.168.1.1? Tell 192.168.1.104
149	83.967909	PCSSystemtec_5e:01:…	PCSSystemtec_b8:b7:…	ARP	42 192.168.1.1 is at 08:00:27:5e:01:7c
210	148.203590	PCSSystemtec_b8:b7:…	PCSSystemtec_5e:01:…	ARP	60 Who has 192.168.1.1? Tell 192.168.1.104
211	148.204596	PCSSystemtec_5e:01:…	PCSSystemtec_b8:b7:…	ARP	42 192.168.1.1 is at 08:00:27:5e:01:7c
305	276.508189	PCSSystemtec_b8:b7:…	PCSSystemtec_5e:01:…	ARP	60 Who has 192.168.1.1? Tell 192.168.1.104
306	276.508569	PCSSystemtec_5e:01:…	PCSSystemtec_b8:b7:…	ARP	42 192.168.1.1 is at 08:00:27:5e:01:7c
340	315.152982	PCSSystemtec_b8:b7:…	PCSSystemtec_5e:01:…	ARP	60 Who has 192.168.1.1? Tell 192.168.1.104
341	315.153278	PCSSystemtec_5e:01:…	PCSSystemtec_b8:b7:…	ARP	42 192.168.1.1 is at 08:00:27:5e:01:7c
3560	481.327624	PCSSystemtec_2d:f8:…	Broadcast	ARP	60 Who has 192.168.1.104? Tell 192.168.1.105

Request



```
▸ Frame 210: 60 bytes on wire (480 bits), 60 bytes captured (480 bits)
▸ Ethernet II, Src: PCSSystemtec_b8:b7:58 (08:00:27:b8:b7:58), Dst: PCSSystemtec_5e:01:7c (08:00:27:5e:01:7c)
▾ Address Resolution Protocol (request)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: request (1)
    Sender MAC address: PCSSystemtec_b8:b7:58 (08:00:27:b8:b7:58)
    Sender IP address: 192.168.1.104
    Target MAC address: 00:00:00_00:00:00 (00:00:00:00:00:00)
    Target IP address: 192.168.1.1
```

Response



```
▸ Frame 211: 42 bytes on wire (336 bits), 42 bytes captured (336 bits)
▸ Ethernet II, Src: PCSSystemtec_5e:01:7c (08:00:27:5e:01:7c), Dst: PCSSystemtec_b8:b7:58 (08:00:27:b8:b7:58)
▾ Address Resolution Protocol (reply)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: reply (2)
    Sender MAC address: PCSSystemtec_5e:01:7c (08:00:27:5e:01:7c)
    Sender IP address: 192.168.1.1
    Target MAC address: PCSSystemtec_b8:b7:58 (08:00:27:b8:b7:58)
    Target IP address: 192.168.1.104
```

1. Identify the ARP request and record its frame number and timestamp.

```
Frame no. 210
Timestamp : 148.203590
```

2. Identify the corresponding ARP reply and record its frame number and timestamp.

```
Frame no. 211
Timestamp : 148.204596
```

3. What is the sender IP address in the ARP request?

```
192.168.1.104
```

4. What is the target IP address being resolved?

```
192.168.1.1
```

5. What is the sender MAC address in the ARP request?

```
08:00:27:b8:b7:58
```

6. What is the target MAC address specified in the ARP request?

```
00:00:00:00:00:00 (sender initailly don;t know the target's physical
address)
```

7. What is the opcode of the ARP request?

```
request (1)
```

8. What are the Hardware Type and Protocol Type?

```
Hardware type : Ethernet (1)
Protocol type : IPV4 (0x0800)
```

9. In the corresponding ARP reply, what MAC address is associated with the requested IP address?

```
Mac Address : 08:00:27:5e:01:7c
```

10. Calculate the time difference between the ARP request and the corresponding ARP reply.

```
Request time : 148.203590 second
Response time : 148.204596 second

Difference : 148.204596 - 148.203590 = 0.001006 seconds
```

11. Explain how the request and reply together establish an IP-to-MAC address mapping.

```
The sender broadcasts an ARP request to the entire local network,
asking for device with IP address 192.168.1.1. The specific device
that owns that IP address processes the request and sends a direct,
unicast ARP reply stating the IP address 192.168.1.1, and the physical
MAC address is 08:00:27:5e:01:7c. The origisnal sender receives this
reply and saves the IP-to-MAC mapping in its local ARP cache so it can
successfully deliver future data frames to that physical address.
```